
DSAN 5300 Quiz-6 Study Guide

Quiz-6 will be posted on Canvas, and will evaluate your understanding of:

- Why we don't want to use **training error** to evaluate our models (since we can always push this down to 0 by increasing the flexibility of our model, but this decreases its **generalizability**),
- Why we can't use **test error** for model *selection* (since the moment any information from the test set “leaks” back into the model-training process, and is used to affect model parameters, it no longer functions as “test” data), and
- How **Cross-Validation** comes to our rescue, allowing us to estimate test error and use these estimates to select an ideal model *before* evaluating this final-choice model on the test set.

Since the last study guide was way too long (and the corresponding quiz was also too different from the material on the guide), here I'm “tying myself to the mast” by **only** including **practice questions** in this guide, several per topic. Then, to ensure that the quiz matches the guide, one question per topic will be chosen at random, removed, and used as the quiz question for that topic!

Topic 1: Training Error and the Scourge of Overfitting

Question 1.1 (Single Answer)

The specific proportions of 80% and 20% are typically used for the train-test split (where we estimate our model parameters on a random sample of 80% of all data, then evaluate the model on the remaining 20%) because...

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- ☐ These values guarantee a lower test error than any other possible split
 - ☐ These values guarantee that the resulting model achieves a mathematically-optimal balance between bias and variance
 - ☐ The ratio of these two values, $\frac{80}{20} = 4$, is the **maximum number of colors required** to color the regions of any map such that no two adjacent regions have the same color.
 - ☐ None of the above: These values tend to perform well in practice, but have no “deeper” theoretical justification
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Solution:

None of the above: These values tend to perform well in practice, but have no “deeper” theoretical justification.

This is important to keep in mind only because, in the history of science, there have been many many “rules of thumb” for “best practices” like the 80-20 split which, though they were considered “common sense” for decades, ended up being sub-optimal — sometimes even harmful! — with respect to the intended goal of the practice¹.

On the other hand, there are *also* prominent practices in science/engineering which were widely used before they were eventually found (via mathematical proof) to in fact be optimal². We just don’t know which of these outcomes lies in store for the 80-20 split, so we keep this warning in the back of our heads just in case!

Question 1.2 (Fill in the Blank)

Given n observations (with distinct x coordinates) $((x_1, y_1), \dots, (x_n, y_n))$ in $\mathbb{R}^2$, we can always construct a unique polynomial of degree ,

$$y = \beta_0 x^0 + \beta_1 x^1 + \beta_2 x^2 + \dots + \beta_{\text{input}} x^{\text{input}}$$

which perfectly predicts y_i from x_i for all i .

Solution:

The number that goes in the blanks here is $n - 1$. For questions like this, rather than thinking in terms of an unspecified n , it can help to choose specific manageable values like 2 or 3, and build intuition from these cases:

When $n = 2$, we have exactly 2 points (with distinct x coordinates) in the xy -plane, meaning that we can always find an intercept value β_0 and a slope value β_1 for the unique line $y = \beta_0 + \beta_1 x$ passing through these two points. In general (as in, for more complicated scenarios with higher n , or in higher dimensions) we could use OLS or maximum likelihood to derive these two parameters, but in this case it’s even easier: if our two points are

$$(x_0, y_0) \text{ and } (x_1, y_1),$$

then we can solve the system of two equations in two unknowns to derive β_0 and β_1 from these two points:

¹Here the obvious example is the $p < 0.05$ cutoff “rule” in statistics... See Ziliak and McCloskey (2008), [The Cult of Statistical Significance: How the Standard Error Costs Us Jobs, Justice, and Lives](#).

²The best example I can think of quickly is *Huffman coding*: Telegraph operators, from the 1830s onwards, “intuitively” developed codes for compressing their messages and/or making error correction easier for their recipients, but it wasn’t until Claude Shannon’s 1948 [source coding theorem](#) that we were able to prove the optimality of perhaps the most prominent among these “intuitively-developed” telegraph codes! See Section 5.8 of Cover and Thomas (2006), [Elements of Information Theory](#).

$$\begin{aligned}y_0 &= \beta_0 + \beta_1 x_0 \\y_1 &= \beta_0 + \beta_1 x_1\end{aligned}$$

Starting with the first equation, we can solve to get $\beta_0 = y_0 - \beta_1 x_0$, and plug this in for β_0 in the second equation:

$$\begin{aligned}y_1 &= y_0 - \beta_1 x_0 + \beta_1 x_1 \\ \Leftrightarrow y_1 - y_0 &= \beta_1 (x_1 - x_0) \\ \Leftrightarrow \beta_1 &= \frac{y_1 - y_0}{x_1 - x_0}\end{aligned}$$

And we can then plug this expression for β_1 back into the first equation to obtain β_0 :

$$\begin{aligned}y_0 &= \beta_0 + \frac{y_1 - y_0}{x_1 - x_0} x_0 \\ \Leftrightarrow \beta_0 &= y_0 - \frac{y_1 - y_0}{x_1 - x_0} x_0 \\ \Leftrightarrow \beta_0 &= \frac{y_0(x_1 - x_0)}{x_1 - x_0} - \frac{y_1 x_0 - y_0 x_0}{x_1 - x_0} \\ \Leftrightarrow \beta_0 &= \frac{y_0 x_1 - y_0 x_0 - y_1 x_0 + y_0 x_0}{x_1 - x_0} \\ \Leftrightarrow \beta_0 &= \frac{y_0 x_1 - y_1 x_0}{x_1 - x_0},\end{aligned}$$

and therefore (though it's a bit messy), the following line in slope-intercept form is the unique line which passes through our two points, so that if we use this line to predict y from x , we achieve an error rate of 0:

$$y = \frac{y_1 - y_0}{x_1 - x_0} x + \frac{y_0 x_1 - y_1 x_0}{x_1 - x_0}$$

If we tried the same thing with $n = 1$, a dataset with only **one** point (x_0, y_0) , there would no longer be a **single** perfectly-fitting line through our points. Instead, there would be **infinitely many** — find one line passing through (x_0, y_0) and then you can rotate it by any amount to derive a new line passing through it — and we'd have no way to use something like RMSE to decide between these lines, since all would achieve an error of zero.

On the other hand, if we tried the same thing with $n = 3$ points, we could no longer guarantee that there exists a line passing through all three points³. This $n > p + 1$ case is less difficult to deal with, however, since now we **can** use a loss function like RMSE to find a line which,

³Such a line would only exist in the “degenerate” case where the three points are collinear.

though it may not **perfectly** fit the data (i.e., it may not perfectly predict y from x), at least “best approximates” it. And... this is exactly what we’re doing when we use OLS or other regression estimation methods to find the “line of best fit” in a regression setting!

Topic 2: Cross-Validation

Question 2.1 (Math-Expression Answer)

Assume you are working with a dataset containing n observations, which you then randomly split into n_{Tr} training observations and $n - n_{\text{Tr}}$ test observations. You are hoping to choose between a third, fourth, and fifth order polynomial regression model, so you decide to use **Leave-One-Out Cross-Validation (LOOCV)** to derive estimates of the test error for each of these models with respect to your dataset.

How many **overlapping observations** are there between any two training “folds” in the LOOCV procedure (that is, any two subsets of the full n_{Tr} training observations which are given to the model at some point in the procedure)? Write your answer in terms of the total number of training observations n_{Tr} .

Solution:

Since the training folds in each step of LOOCV include **every** training observation besides the single held-out observation i (which forms the single-element “validation set”), any two such training folds will have exactly $\boxed{n_{\text{Tr}} - 1}$ overlapping observations. As the book puts it on page 209, this leads to LOOCV having higher variance than k -Fold CV with $k < n$, because:

When we perform LOOCV, we are in effect averaging the outputs of n fitted models, each of which is trained on an almost identical set of observations; therefore, these outputs are highly (positively) correlated with each other.

In contrast, when we perform k -fold CV with $k < n$, we are averaging the outputs of k fitted models that are somewhat less correlated with each other, since the overlap between the training sets in each model is smaller.

Since the mean of many highly correlated quantities has higher variance than does the mean of many quantities that are not as highly correlated, the test error estimate resulting from LOOCV tends to have higher variance than does the test error estimate resulting from k -fold CV.

Question 2.2 (Fill in the Blank)

When estimating test error using n_{Tr} training observations, K -Fold Cross-Validation is equivalent to LOOCV when $K =$ _____

Solution:

$K = n_{\text{Tr}}$. Since each step in LOOCV (each individual estimate that is averaged at the end to produce the overall LOOCV estimate of test error) trains the model on $n_{\text{Tr}} - 1$ observations and evaluates this trained model on its ability to predict the single remaining non-included observation, by the end of the procedure (when the overall mean of the individual estimates is computed) each of the n_{Tr} points has served as the held-out point once, thus resulting in n_{Tr} separate folds being used in the eventual estimate.

Topic 3: CV and the Bias-Variance Tradeoff

For the questions in this topic, consider the three forms of Cross-Validation we've learned:

- The **Validation Set Approach**
 - (A single estimate of test error, derived by training the model on some proportion p of the n_{Tr} total training observations, then estimating test error by evaluating the model on the remaining $n - n_{\text{Tr}}$ observations)
- **5-Fold Cross-Validation**
 - (The average of $K = 5$ estimates of test error, derived by splitting the full set of training observations randomly into 5 subsets of approximately equal size, then treating each subset as the held-out validation set once and as part of the model's training data the other four times)
- **LOOCV**
 - (The average of $K = n_{\text{Tr}}$ estimates of test error, derived by treating each observation i in the training data as the single held-out point once, and as part of the model's training data the other $n_{\text{Tr}} - 1$ times)

Question 3.1 (Fill in the Blanks, with Word Bank)

Fill in the blanks in the following inequality based on where the three methods rank with respect to **bias** when used to estimate test error (for brevity, you can shorten the names of the methods to "VSA", "5-Fold CV", and "LOOCV"):

$$\text{Bias}^2[\quad] < \text{Bias}^2[\quad] < \text{Bias}^2[\quad]$$

Solution:

As discussed on pages 208 to 209 in ISLP (better than I could summarize it!),

$$\text{Bias}^2[\text{LOOCV}] < \text{Bias}^2[k\text{-Fold CV}] < \text{Bias}^2[\text{VSA}]$$

Question 3.2 (Fill in the Blanks, with Word Bank)

Fill in the blanks in the following inequality based on where the three methods rank with respect to **variance** when used to estimate test error (for brevity, you can shorten the names of the methods to “VSA”, “5-Fold CV”, and “LOOCV”):

$$\text{Var}[\quad] < \text{Var}[\quad] < \text{Var}[\quad]$$

Solution:

In this case, the quoted portion of page 209 above contains the explanation for the following ranking:

$$\text{Var}[\text{VSA}] < \text{Var}[k\text{-Fold CV}] < \text{Var}[\text{LOOCV}]$$

Question 3.3 (Single Answer from Word Bank)

Given your responses to Question 3.1 and Question 3.2, which of the three methods likely does best in terms of achieving an effective balance between **bias** and **variance**?

Solution:

5-Fold Cross-Validation! One can prove⁴ that, under mild assumptions, K -fold CV with “reasonable” values of K like $K = 5$ or $K = 10$ will outperform both the Validation Set approach and (almost always) LOOCV. Much like the use of the 80-20 split discussed above, however, the particular values within this range of “reasonable” values that people choose (almost always $K = 5$ or $K = 10$) are driven more by convention than by statistical principle. As mentioned in the footnote, you *can* dig into ESL references to eventually find a proof that there **is** in fact a theoretical optimal K value, but it turns out that the performance differences you may be able to achieve by “optimizing” K in this way are usually not worth the additional effort, relative to the ease of adhering to the $K = 5$ or $K = 10$ “rule of thumb” and moving on with your project!

⁴A more in-depth analysis is given in *Elements of Statistical Learning*, the fancier but way longer version of ISLP, but for the full-on proof, you need to dig into its references, which will eventually lead you to Shao (1997), “An Asymptotic Theory for Linear Model Selection”.